

Redmi Note 5 Level 3 Maintenance Guide V01

Technical support internal document control: TSIMHNE7S Redmi Note 5 Level 3 Maintenance Guidance V01

Scope of application:

Analysis center, mainboard and whole machine maintenance factory

Change history:

First edition 2018/6/8

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1. Introduction of basic information

1.1 Product Overview

Processor and memory:

骁龙636八核处理器

骁龙636处理器，最高主频 1.8GHz

Adreno™ 509 图形处理器

6GB 大内存最高可选

3GB / 4GB / 6GB LPDDR4X 1333MHz 双通道

可选 32GB / 64GB eMMC 5.1 高速闪存

Camera:

后置双摄

AI 双摄

1200万+500万 旗舰双摄相机

1.4 μ m 超大像素主摄，暗光拍照更清晰

Dual PD 双核极速对焦

f/1.9 大光圈镜头
AI 人像模式

暗光增强技术

高动态范围调节技术

全景模式

连拍模式

面部识别功能

All Netcom 5.0:

Support for dual cards is not limited to operators, and can be stationed in the network with 4G (Note 1: Whether to support 4G depends on the local operator network)

Double Nano-SIM card slots, any card slot can be set as the main card, supporting mobile and Unicom. Telecom 4G/3G/2G supports mobile and Unicom in-band carrier aggregation 4G + network

Support dual-card VoLTE HD voice (Note 2: Limited by operator network and service deployment, only China Mobile VoLTE service is currently supported, and other operators' VoLTE requirements can be supported through software upgrade in the future)

详细网络频段

2G:GSM (频段 B2/3/5/8)

2G:CDMA (频段 BC0)

3G:CDMA EVDO (频段 BC0)

3G:WCDMA (频段B1/2/5/8)

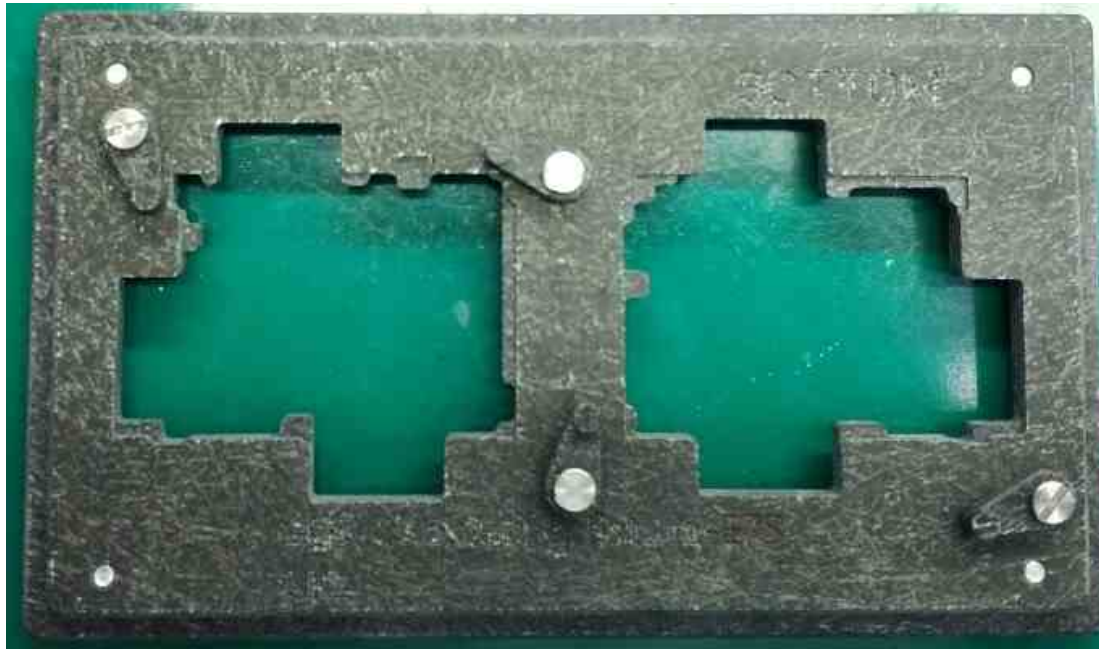
3G:TD-SCDMA (频段 B34/39)

4G:TDD-LTE (频段 B34/B38/39/40/41 (100MHz))

4G:FDD-LTE (频段 B1/3/5/7/8)

1.2 Red Rice Note 5 Special Welding Fixture

Material code of welding fixture: SCNC020042800



1.3 Redmi Note 5 power supply adapter

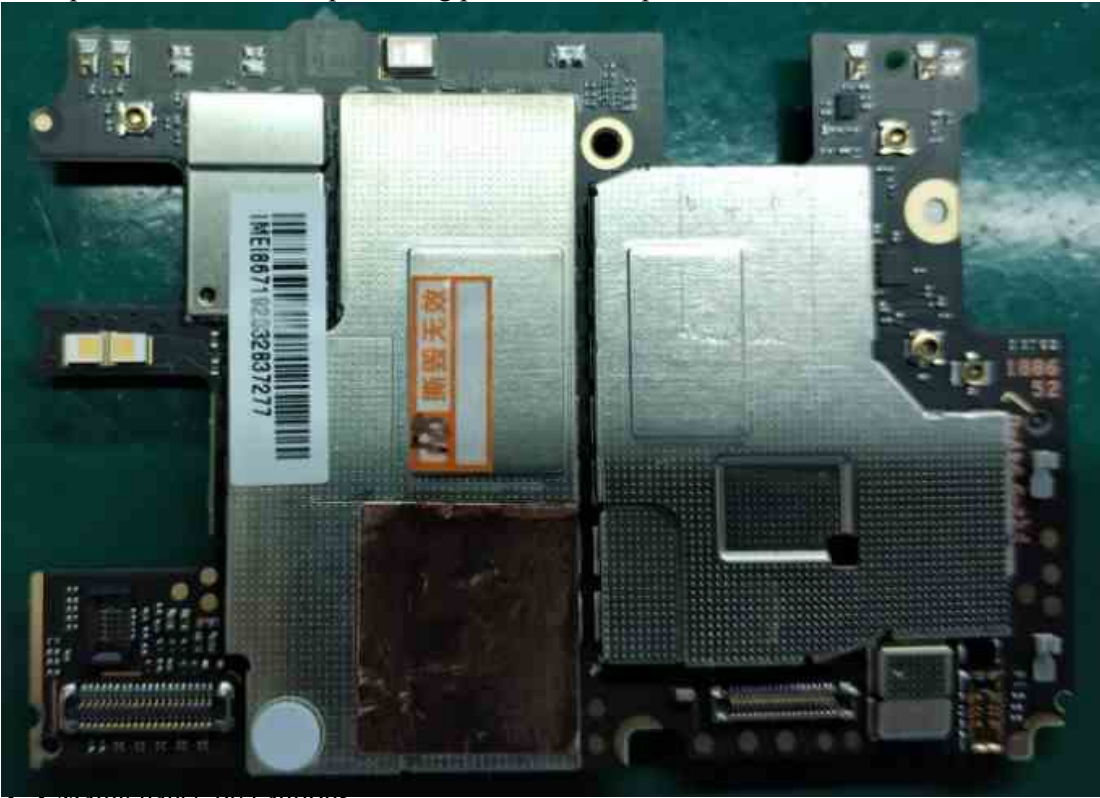
Material code: SCNC030037300



1.4 Pasting position and specification of maintenance label

Repair label: 12mm*5mm

Paste position: Paste on the protruding position on the power shield.



1.5 Maintenance precautions

1. Material change:

B903 Material Change: B903 Both primary and secondary materials can be used

23C000RNA002	RC0402JR-070RL	RES,0R,YAGEO RC0402JR-070RL,EIA0402,50V,±5%	YAGEO 一供
23B0000MA025	0402WGJ0000TCE	RES,0402SMD Resistor(0Ω ±5% 1/16w)	UNIOHM 二供

L906 Material Change: L906 Material only uses secondary materials

24PM184R7093	WPN252010HS4R7MT	Ind,4.7 μ H, $\pm$ 20%,2A,0.13 Ω ,2520,1mm	SUNLORD 二供
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For the motherboard with black screen fault of this model, the fault points are mostly concentrated on the original B902, B903 and L906 of VREG LEDA booster circuit, focusing on measuring these components. If B903 is not conductive, replace B903 and L906 at the same time; If the L906 element is not conductive and B903 is normal, only L906 can be replaced.

2. CPU and EMMC replacement rules

The DDR of Redmi Note 5 is integrated with eMMC, and the CPU is bound with EMMC, so EMMC can be replaced separately. However, to replace CPU, EMMC must be replaced. If key is not bound, only CPU needs to be replaced.

3. When CIT tests the charging function of Red Rice Note 5, it needs to plug in the charger and connect the USB cable of the desktop computer to pass.

4. Pay attention to protect plastic components when welding the motherboard to avoid welding.

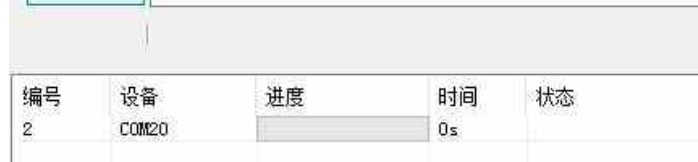
1.6 MiFlash brush

- Turn off the phone, make the phone enter deep brush mode (identify to port 9008): short-circuit TP22, TP23 and then connect USB cable



- Qualcomm HS-USB QDLoader 9008 (COM20)

- Click on the MiFlash tool "Load Device" button, read COM port, click on the "Brush Machine" button, start brushing machine.



编号	设备	进度	时间	状态
2	COM20		0s	

Clean NV factory package: E7SCN_EDL_eraseNV_LLDLME5_2017.
SPF.2.1_0.0012.0_T16C0207

Retain NV factory package: E7SCN_EDL_eraseNV_LLDLME5_2017.
SPF.2.1_0.0.0012.0_T16C0207

1.7 RF calibration test correlation

- Brush the NV factory package deeply and write it to FSN with DT tool.
- Open the calibration software, and connect the main antenna and the auxiliary receiving RF line when the mobile phone is turned on. Connect
- the USB cable, and the mobile phone starts RF calibration.
- After calibration, if you want to test the signal function, you need to use DT tool to restart the mobile phone, otherwise there is no signal, and then brush MIUI software for function test.

2. Brief introduction of motherboard module

2.1 Brief introduction of red rice Note 5 startup timing and key signal measurement table

Boot Timing:

开机时序测量表		
Symbol	测量值	测量点
KYPD_PWR_N	1.8V	TP9
VREG_BOB	3.3V	C530
VREG_S5B_0P915	0.915V	L905
VREG_S3B_S4B	0.87V	L903
VREG_S2B	1.05V	C925
VREG_L3A_1P0	1V	C512
VREG_S4A	2.04V	C606
VREG_S5A	1.35V	C609
VREG_L13A_1P8	1.8V	C525
VREG_L10A_1P8	1.8V	C515
VREG_L9A_1P8	1.8V	C523
VREG_L6A_1P3	1.3V	C513
VREG_L11A_1P8	1.8V	C524
REF_MSM_HVPAD	1.25V	C411
VREG_L10B_0P915	0.915V	C817
VREG_L9B_0P87	0.87V	C816
VREG_L1B_0P925	0.925V	C808
VREG_L7B_3P125	3.125V	C811
VREG_L2B_SDC2	2.95V	C812
VREG_L5B_2P95	2.95V	C810
VREG_S1B_1P125	1.125V	L901
VREG_L4B_2P95	1.95V	C813
VREG_L8A_1P8	1.8V	TP23
SLEEP_CLK	32KHz	R512
VREG_S1A_0P87	0.87V	L601
VREG_L14A_1P8	1.8V	C517
VREG_L1A_1P225	1.225V	C511
LNBBCLK	19.2MHz	R507
PON_RESET_N	1.8V	R202
SDM_PS_HOLD	1.8V	R503

CPU (U201) power supply signal and measuring point, the yellow part is in the startup timing signal:

CPU供电测量表		
Symbol	测量值	测量点
VREG_S5B_0P915	0.915V	L905
VREG_L10B_0P91	1.8V	C515
VREG_L10A_1P8	1.8V	C515
REF_MSM_HVPAD	1.25V	C411
VREG_L2B_SDC2	2.95V	C812
VREG_S1A_0P87	0.87V	L601
VREG_S3B_S4B	0.87V	L903
VREG_L9B_0P87	0.87V	C816
VREG_L1B_0P925	0.925V	C808
VREG_L1A_1P225	1.225V	C511
VREG_L7B_3P125	3.125V	C811
VREG_L13A_1P8	1.8V	C525
VSIM1	C1313 (SIM1插卡)	2.9V
VSIM2	C1314 (SIM2插卡)	2.9V
VREG_L6A_1P3	1.3V	C513
VREG_L9A_1P8	1.8V	C523
VSNS_VDDIO_EBIX	0.6V	R402
VREG_S6A	0.7V	C430
VREG_L5A_0P848	0.848V	C521
VREG_S1B_1P125	1.125V	L901
VREG_S2A_S3A_0P87	0.87V	L602

PM660:

11	33	55	77	99	120	142	164	186	208
VSW_S2	VDD_S2	VDD_S2	VSW_S4	VSW_S6	VDD_S6	VSW_S5	GND_S5	VDD_S4	VDD_S4
22	44	66	87	109	131	153	175	197	219
VSW_S2	VSW_S2	VDD_S1	GND_S1	VSW_S6	VDD_S5	VSW_S5	VDD_S4	VSW_S4	VSW_S4
10	32	54	76	97	119	141	163	185	207
GND_S2	GND_S2	VSW_S3	VSW_S1	GND_S6	VREG_S6	VDD_S5	GND_S5	VSW_S4	GND_S4
21	43	65	86	108	130	152	174	196	218
GND_S3	GND_S3	VDD_S1	GND_S1	SPHIL_CLK	VPH_PN_R_1	PON_OUT	VREG_S4	GND_S4	CBL_PWR_M
9	31	53	75	96	118	140	162	184	206
VSW_S3	VSW_S3	VSW_S3	VSW_S1	VREG_S1	SPHIL_DATA	VREG_BOB	VREG_S5	RESIN_N	PS_HOLD
20	42	64	85	107	129	151	173	195	217
VDD_S3	VDD_S3	VDD_S3	VDD_S1	VREF_NEG_S1	NC	VREF_MSM	AVDD_BYP	PON_RESET_N	FAULT_M
8	30	52	74	95	117	139	161	183	205
GPIO_2	GPIO_8	GPIO_11	VREG_S3	VREG_S2	GPIO_13	VDD_HSM_IQ	AMUX_1	ANA_IN	VREG_L_1
19	41	63	85	106	128	150	172	194	216
GPIO_7	GPIO_10	GPIO_3	VREF_NEG_S3	GND	SLEEP_CLK	AMUX_5	AMUX_3	VDD_L1_6_7	VREG_L7
7	29	51	73	94	116	138	160	182	204
GPIO_5	GPIO_9	GPIO_3	VREF_NEG_S3	GND	GND	GND	AMUX_3	RD_THERM	VDD_L1_6_7
18	40	62	84	105	127	149	171	193	215
VSW_HAP_P	VSW_HAP_M	GPIO_12	GND_HAP	GND	GND	TEST_EN_VPP	AMUX_4	GND_XD_ADC	VREG_L6
6	28	50	72	93	115	137	159	181	203
VDD_HAP	PGND_HAP	HAP_PWM_IN	VDD_PORHY	GND	GND_PORHY	BB_CLK_3	GPIO_6	VREG_L14	VREG_L5
17	39	61	83	104	126	148	170	192	214
VCONNL_IN	CC_OUT	VCONNL_EN	GND	GND_WLP_TST	GND	BB_CLK_1	NC	VREG_L9	VDD_L5
5	27	49	71	92	114	136	158	180	202
ECCL_ID	CC2	USB_DP	IPO_PWR_3	WIPWR_RECHG	GND	BB_CLK_3	BB_CLK_L1H	REF_BYP	VREG_L8
16	38	60	82	103	125	147	169	191	213
VBATT_PWR	VBATT_PWR	USB_DM	GND_CHG	Q1_PMA_ON	GND	GPIO_4	GND_REF	VDD_L1H_2_12_14	VDD_L1H_2_12_14
4	26	48	70	91	113	135	157	179	201
VBATT_PWR	VBATT_PWR	VBATT_PWR	GND_CHG	WIPWR_CHG_OK	BATT_ID	GND	BA_N	VREG_L13	VREG_L12
15	37	59	81	102	124	146	168	190	212
VPH_PWR	VPH_PWR	REF_GND_CHG	VBATT_SNS_M	VBATT_SNS_P	GND_PSUB_FG	RF_CLK1	RF_CLK2	VREG_L10	GND_RF
3	25	47	69	90	112	134	156	178	200
VPH_PWR	VPH_PWR	VPH_PWR	GND_CHG	STAT_CHG	VARS	GND_FG	BATT_THERM_BIAS	VREG_L11	VREG_RF_CLK
14	36	58	80	101	123	145	167	189	211
BOOT_PWR	BOOT_CAP	GND_CHG	DC_SNS	USB_SNS	VREG_FG	AUX_THERM_BIAS	REF_GND_FG	VDD_XD_RFCLK	VREG_XD
2	24	46	68	89	111	133	155	177	199
LDQ_CTRL	PGND_CHG	VSW_CHG	USB_IN_MID	USB_IN	VREG_L18	VBATT_SNS_M	ISNS_SNS_M	AUX_THERM	GND_XD
13	35	57	79	100	122	144	166	188	210
PGND_CHG	VSW_CHG	USB_IN_MID	USB_IN	USB_EN	VBATT_SNS_P	ISNS_SNS_P	BATT_THERM	GND_XD_150	STAT_IN
1	23	45	67	88	110	132	154	176	198
LDQ_CTRL	PGND_CHG	VSW_CHG	USB_IN_MID	USB_IN	VREG_L15	VDD_L1H_16_17_18_19	VREG_L17	VREG_L3	XTAL_OUT
12	34	56	78	99	121	143	165	187	209
PGND_CHG	VSW_CHG	USB_IN_MID	USB_IN	DC_EN	VREG_L19	VREG_L16	VDD_L2_3	VREG_L2	GND_XD

PM660L:

	22		44		66		88		110		130		150		174		196
	VSW_O1 SP_M		VDISP_1 M_OUT		PGND_0 ISP		VDISP_P _MID		PGND_0 LEDB		VREG_O LEDB		VREG_B DOST		VSW_BO OST		GND_BO OST
11		33		55		77		99		121		141		163		185	
VDD_O1 SP_M		VSW_O1 SP_M		VDISP_P _FB		VSW_O1 SP_P		VDISP_P _OUT		VSW_O1 EDB		CDC_P0 M_RX2		GND_BO OST		VDD_CP	
	21		43		65		87		109		129		151		173		195
	VDD_O1 SP_M		NC		VDD_O1 SP_P		VDISP_1 DO_IN		NC		VREG_O LEDB_FB		CDC_P0 MLTX		CP_CL_P		GND_CP
10		32		54		76		98		120		140		162		184	
GNP_SC TRL		VDISP_1 M_FB		GND_O1 SP_M		GND_O1 SP_P		GND_O1 EDB		NC		CDC_P0 M_CLK		BOOST_1 SNS		CP_CL_1 N	
	20		42		64		86		108		128		150		172		194
	VREG_B OE		VREG_B OB		SPM_CL_1 N		OLED_B_1 EN		VDD_O1 EDB		NC		CDC_P0 M_RX0_1 DBE		GND_SP KR_PA		CP_VME G
9		31		53		75		97		119		139		161		183	
VSW_BS T_BOB		VSW_BS T_BOB		GND_PS UB_BOB		SPM_O_1 KTA		GPIO_07		CDC_P0 M_RX3_1 ONE		CDC_P0 M_RX1		CDC_P0 M_SYNC		SPM_O_1 RV_M	
	19		41		63		85		107		127		149		171		193
	VSW_BS T_BOB		VREG_B OB_SNS		GPIO_10		GND		WUP_TS T		CDC_P0 M_RX0		VDD_AU DIO_00		GND_SP KR_PA		SPM_O_1 RV_P
8		30		52		74		96		118		138		160		182	
PGND_B OB		PGND_B OB		VREG_1 B		VPM_PW RL_1		ISND		GND		MIC_IN1		MIC_IN2		VDD_SP KR_PA	
	18		40		62		84		106		126		148		170		192
	VSW_BC K_BOB		EXT_CT RL_BOB		VDD_L4 _B		GND		GND		MIC_IN1 _M		HS_DET		NC		VDD_SP KR_PA
7		29		51		73		95		117		137		159		181	
VSW_BC K_BOB		VSW_BC K_BOB		VREG_1 4		VDD_RG B		GND		GND		MIC_IN1 _P		EARD_M		EARD_P	
	17		39		61		83		105		125		147		169		191
	VDD_BO B		VDD_BO B		RGB_RE D		GPIO_06		GND				MIC_BIA S2		VDD_HP H		HPH_RE
6		28		50		72		94		116		136		158		180	
BA_N		PGN_1		RGB_GR N		RGB_BL U		TEST_SH _VPP		GND		GND_CP ILT		MIC_BIA S1		VREG_H PH	
	16		38		60		82		104		124		146		168		190
	FLASH_S TROBE		GND_FL ASH		REF_BYP		GND		GND		GND		GPIO_03		HPH_RE F		HPH_L
5		27		49		71		93		115		135		157		179	
FLASH_L ED1		GND_FL ASH		DIE_TEMP		REF_GND D		VREG_S 3		VREF_N EQ_S1		VREG_S 1		GPIO_11		GPIO_05	
	15		37		59		81		103		123		145		167		189
	FLASH_L ED2		FLASH_L ED3		VREF_N EQ_S2		VREF_N EQ_S3		VREF_N EQ_S4				GPIO_02		GPIO_08		GPIO_09
4		26		48		70		92		114		134		156		178	
VDD_FL ASH		VDD_FL ASH		AVDD_B YP		VREG_S 2		NC		VREG_S 4		GPIO_04		VDD_S1		VDD_S1	
	14		36		58		80		102		122		144		166		188
	VREG_L 1		VREG_L 2		VPM_PW RL_1		FAULT_N		GPIO_01		VREG_S 5		VSW_S1		VSW_S1		VSW_S1
3		25		47		69		91		113		133		155		177	
VDD_L1 _9_10		VDD_L2		GND_S3		VSW_S3		VSW_S4		VDD_HS MLD		GPIO_12		GND_S1		GND_S1	
	13		35		57		79		101		121		143		165		187
	VREG_L 10		VREG_L 9		VSW_S3		GND_S4		VDD_S4		VREG_L 3		VREG_L 7		VSW_S5		VDD_S5
2		24		46		68		90		112		132		154		176	
VDD_S1		VDD_S2		GND_S3		VDD_S3		VSW_S4		VDD_L3 _5_7_8		VREF_N EQ_S5		GND_S5		VDD_S5	
	12		34		56		78		100		120		142		164		186
	VSW_S2		VSW_S2		VSW_S3		GND_S4		VDD_S4		VREG_L 8		VDD_L3 _5_7_8		VSW_S5		VDD_S5
1		23		45		67		89		111		131		153		175	
VSW_S2		GND_S2		GND_S2		VDD_S3		VSW_S4		VDD_S4		VREG_L 5		GND_S5		VSW_S5	

3.1 Switch-on and shutdown faults

In the process of maintenance without starting up, we should follow the principle of software before hardware, pay attention to observe whether the motherboard components are damaged, breakdown, liquid inflow, etc. In the specific measurement, we should measure in turn according to the sequence of starting up.

3.1.1 Constant current does not start up

Maintenance analysis ideas: 1. Judge whether the motherboard is at high startup current. If it is at high startup current (the startup current is greater than the good motherboard in the same state), measure whether the output voltage of power supply (U501, U801) is normal.

2. Software upgrade and troubleshooting.

3. If the brush machine reports an error, measure whether the working condition of U1001 is normal.

Maintenance Case 1

Fault phenomenon: No startup,
129mA constant current fault
component: U1001

Maintenance analysis: No startup, 129mA constant current, brush error, replace U1001, brush machine, startup is normal.

Maintenance Case 2

Fault phenomenon: No startup,
160mA constant current fault
component: U201

Maintenance analysis: No startup, 160mA constant current, brushing error, replacing U1001, the fault remains the same, replacing U201, brushing machine, startup is normal.

Maintenance Case 3

Fault phenomenon: No startup,
90mA constant current fault cause:
software upgrade

Maintenance analysis: No startup, 90mA constant current, fault repair after brushing the machine.

3.1.2 The startup current is not maintained

Maintenance analysis ideas:

1. Software upgrade and troubleshooting.

2. Measure whether the startup timing signal is normal and whether there is short circuit.

3. Measure whether the power supply of U501 and U801 is normal.

Maintenance Case 1

Fault phenomenon: No startup,
power failure of white rice Fault
component: U501

Maintenance analysis: When the white rice is powered down, it is found that the power supply U501 is abnormally hot, and the fault is repaired after replacement.

Maintenance Case 2

Failure Phenomenon: No boot, 150mA
no maintenance Faulty Component:
U501

Maintenance analysis: No startup, 150mA is not maintained, short circuit of VREG_S4A is measured, short circuit is no longer removed after U501, and fault is repaired after replacement.

Maintenance Case 3

Failure Phenomenon: No boot, 90mA
does not maintain faulty component:
C803

Maintenance analysis: No startup, 90mA is not maintained, VREG_BOB short circuit is measured, C803 components of motherboard are observed to be abnormal, no short circuit is stopped after removal, and the fault is repaired after replacement.

Maintenance Case 4

Failure Phenomenon: No boot, 63mA
does not maintain faulty component:
U201

Maintenance analysis: No startup, 63mA is not maintained, short circuit of VREG_S4A is measured, short circuit is no longer removed after U201, and fault is repaired after replacement.

3.1.3 No current for startup

Maintenance analysis ideas:

1. Check whether the appearance of J1301 and J1302 is damaged. If the interface and shrapnel are normal, measure whether the ground value of J1301 is normal (BATT_ID, BATT_THERM, VBATT).
2. Power-on measurement VPH_PWR output is normal, KYPDPWR_N (R1304) 1.8 V voltage is normal.
3. Measure whether the startup timing signal is normal.

3.1.4 Leakage does not start up

Maintenance analysis ideas: 1. First of all, visually check whether there are components rupture, breakdown, liquid corrosion and discoloration in the appearance of the motherboard.

2. Measure the value of VBATT and VPH_PWR to ground and whether the voltage is normal. If it is normal, power up and find the heating element.
3. Measure whether the output signal circuit of power supply (U501, U801) is normal.

Maintenance Case 1

Fault phenomenon: No startup,
leakage 2.5 A Fault component:
C611

Maintenance analysis: Do not start up, leakage 2.5 A, measure VPH short circuit, observe C611 discoloration, no longer short circuit after removal, repair after replacement.

Maintenance Case 2

Fault phenomenon: No startup,
leakage 350mA Fault component:
U501

Maintenance analysis: No startup, leakage 350mA, short circuit of measuring VPH, serious heating of power supply U501, no short circuit after removal, fault repair after replacement.

Maintenance Case 3

Fault phenomenon: No startup,
leakage 2.5 A Fault components:
U201, U501

Maintenance analysis: No startup, leakage 2.5 A, short circuit measurement of VPH, DM, DP and VBUS, no short circuit of VPH and VBUS after removing U501, no short circuit of DM and DP after removing U201, and fault repair after replacement.

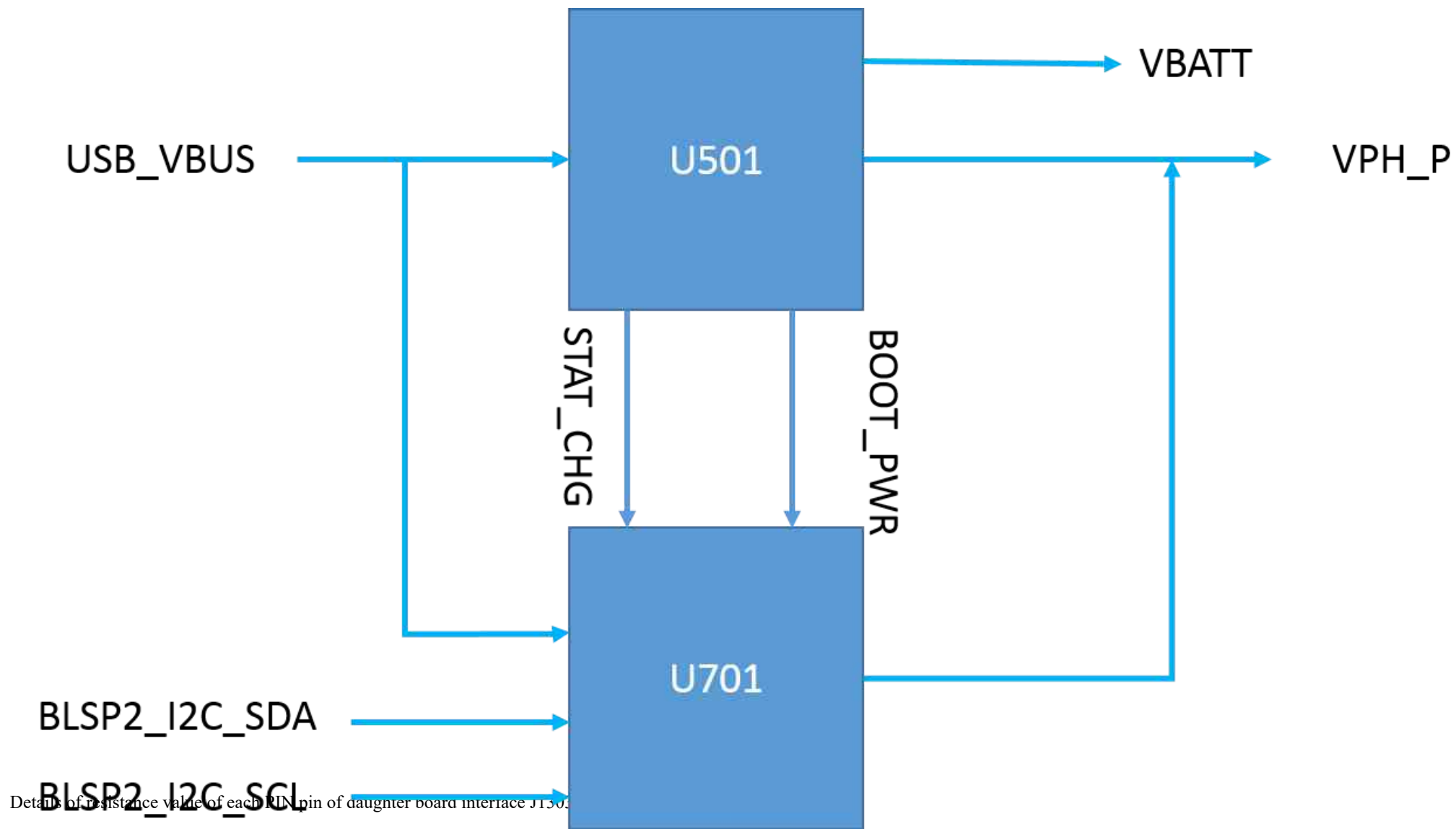
Maintenance Case 4

Fault Phenomenon: High Power
Current 600mA Fault
Components: U201, U801

Maintenance analysis: the startup current is 600mA, and the short circuit of VREG_S3B_S4B is measured. After removing U201, the short circuit is no longer stopped. After replacement, the startup current is 800mA, and U801 is found to be abnormally hot. After replacement, the fault is repaired.

3.2 Charging function failure

U501 and U701 are used for charging, and VBUS enters U501 and U701 respectively. Charging schematic diagram:



		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19		
		430	420	420	425	630	500	GND	600	500	GND	340	340	GND	739	600	600	600	600	GND		
39	GND	J1303																			545	42
40	GND																				545	41
		430	GND	∞	715	590	850	GND	600	600	600	GND	823	GND	740	600	600	600	600	GND		
		38	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20		

Maintenance analysis ideas: 1.

Software upgrade,
troubleshooting software.

2. Check whether the appearance of J1303 and J1301 is normal, and measure whether the four groups of signals USB3_VBUS, USB3_HS_DP, USB3_HS_DM and USB3_HS_ID are normal to ground with multimeter diode gear.

3. Measure whether the VBATT, BATT_ID and BATT_THERM of J1301 are normal to ground.

3. Measure whether the working conditions of U501 and U701 are normal.

4. To ensure that the above signals are normal, consider replacing U501/U701.

Maintenance Case 1

Fault Phenomena: Charging and not
charging. Faulty Component: U701

Maintenance analysis: sometimes charging and sometimes not charging, and repair the fault after replacing U701.

Maintenance Case 2

Fault Phenomenon:
Charging Current Small
Fault Component: L701

Maintenance analysis: Charging current is small, replace L701 fault repair.

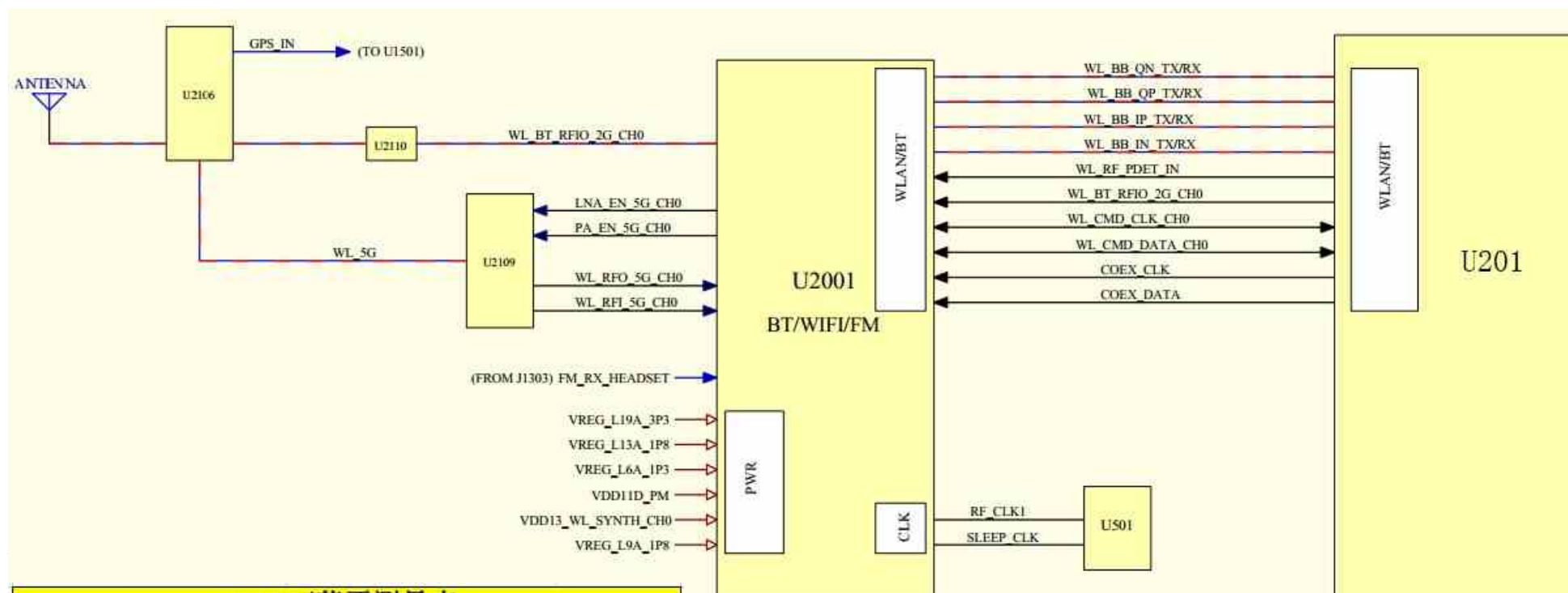
Maintenance Case 3

Fault Phenomenon:
Charging Current Small
Fault Component: L607

Maintenance analysis: Charging current is small, replace L607 fault repair.

3.3 WIFI/BT function failure

Principle block diagram:



WIFI/蓝牙测量表		
Symbol	测量值	测量点
VREG_L19A_3P3	3.3V	C2012
VREG_L13A_1P8	1.8V	1.8V
VREG_L6A_1P3	1.3V (打开蓝牙或WIFI)	C2023
VDD11D_PM	1V (打开蓝牙或WIFI)	C2001
VDD13_WL_SYNTH_CH0	1.3V (打开蓝牙或WIFI)	C2009
VREG_L9A_1P8	1.8V (打开蓝牙或WIFI)	C2017
SLEEP_CLK	32KHz	R512

Maintenance analysis ideas:

1. Software upgrade and troubleshooting.
2. Measure whether CLK is normal.
3. Measure whether the power supply, clock and enable signals of U2001 are normal.

4. Take off U2001 and measure whether the bus between U201 is normal. If it is normal, replace U2001.

5. Replace U201.

Maintenance Case 1

Fault phenomenon: WIFI can't be opened, baseband unknown fault
component: U201

Maintenance analysis: WIFI can't be opened, baseband is unknown, and the fault is repaired after replacing U201.

Maintenance Case 2

Failure phenomenon:
WIFI can't be opened.
Failure reason: software
upgrade

Maintenance analysis: Detect that WIFI can't be opened, brush the NV factory package, and repair the fault.

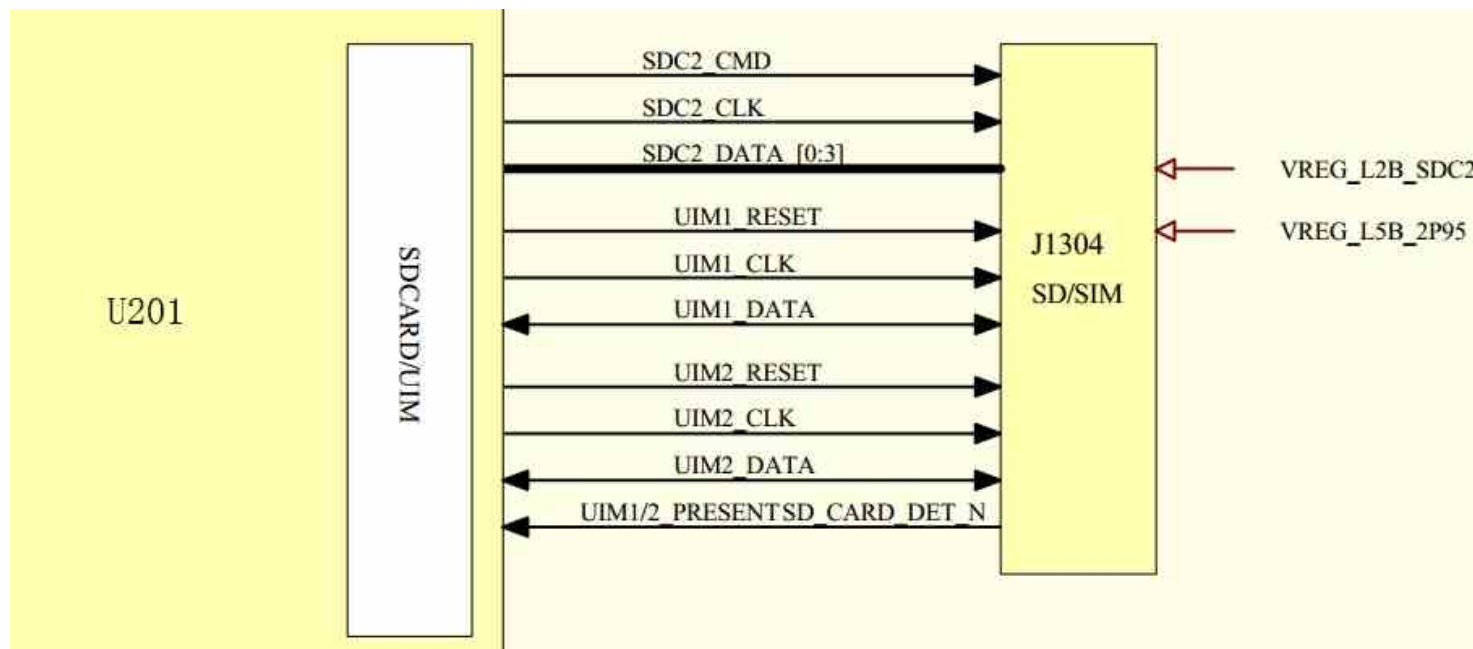
3.4 Function failure of SIM/SD card

SIM card signal control chart:

SIM/SD卡测量表		
Symbol	测量值	测量点
SDC2_DATA_1	410	J1304第14PIN
SDC2_DATA_0	410	J1304第13PIN
SDC2_CLK	410	J1304第11PIN
VREG_L5B_2P95	2.95V	C1316
SDC2_CMD	410	J1304第9PIN
SDC2_DATA_3	410	J1304第8PIN
SDC2_DATA_2	410	J1304第7PIN
VSIM2	648	C1314
UIM2_RESET	420	J1304第17PIN
UIM2_DATA	420	J1304第16PIN
UIM2_CLK	420	J1304第15PIN
VSIM1	648	C1313
UIM1_RESET	420	J1304第3PIN
UIM1_DATA	420	J1304第2PIN
UIM1_CLK	420	J1304第1PIN
VREG_L13A_1P8	1.8V	R1322
SD_CARD_DET_N	1.8V (插卡托)	R1315
UIM1_PRESENT	1.8V (插卡托)	R1317
UIM2_PRESENT	1.8V (插卡托)	R1316



Principle block diagram of SIM/SD card:



Maintenance analysis ideas: 1. First, check whether the baseband version of the mobile phone is normal. If the baseband information is normal, it is the SIM card function failure.

2. Check whether the SIM/SD card pin is deformed, oxidized or broken, and carefully observe whether there is virtual welding phenomenon in the solder joints of SIM/SD card holder.
3. Measure whether the SIM/SD card targets the ground value normally.
4. Start up and test whether the voltage of power supply, clock, reset and data signal of SIM/SD card is normal.
5. If the above signals are normal, replace U201.

3.5 Restart type failure

Maintenance analysis ideas:

1. Check whether the KYPDPOWER_N 1.8 V voltage is pulled low.
2. Software upgrade, troubleshooting software.
3. Measure whether the I2C circuit is normal.
4. Measure whether there is short circuit in the output of U501 and U801, and focus on checking the power supply of U2001/U1501.
5. The software upgrade is invalid. Measure whether U201 power supply and clock are normal.
6. Replace U201/U1001

Maintenance Case 1

Failure phenomenon: boot Android restart

Faulty component: U201

Maintenance analysis: Start Android and restart, and repair the fault after replacing U201.

3.6 Signal function failure

Maintenance analysis ideas:

1. Insert SIM card to ensure normal recognition, and eliminate faults that do not recognize SIM card.
2. Software upgrade, troubleshooting software.
3. RF calibration, check the specific test items through the test report. However, measure the RF path according to the corresponding system and principle block diagram to find the fault point.
4. Measure whether the power supply of RF circuit is normal, and whether the RFFE (1/2/4) CLK/DATA is normal.
5. If the RF calibration is normal, there is still no signal, check whether the SIM card circuit (focusing on measuring whether the DATA signal is normal) is normal.

3.7 Inductor function failure

Redmi Note 5 has U1401 (M-SENSOR), U1402 (P-SENSOR), U1403 (G + G) and J1303 (HALL interface)

Distance sensors, gravity sensors, gyroscopes and compasses share the same set of I2C.

Maintenance analysis ideas: 1.

Software upgrade,
troubleshooting software.

2. Measure whether the working conditions of the corresponding sensors are normal.
3. Replace the corresponding sensor elements.
4. Replace U201.

3.8 Crash type fault

Maintenance analysis ideas:

1. Software upgrade and troubleshooting.
2. Measure whether BAT_ID and BAT_THERM are normal.
3. Measure whether the working conditions of U1001 are normal and replace U1001.
4. Measure whether the power supply of U201 is normal and replace U201.

Maintenance Case 1

Failure Phenomenon:

Application Crash

Failure Cause:

Software Upgrade

Maintenance analysis: application crash, detection stress test crash, software upgrade, fault repair.

Maintenance Case 2

Fault phenomenon: Camera jammed

Cause of failure: Software upgrade

Maintenance analysis: camera application jamming, software upgrade, fault repair.

Maintenance Case 3

Fault Phenomenon: Start-

up White Rice Fixed

Screen Fault Component:

U201

Maintenance analysis: Start the white rice screen, replace U201, and repair the fault.

Maintenance Case 4

Fault Phenomenon:

Application of Crash

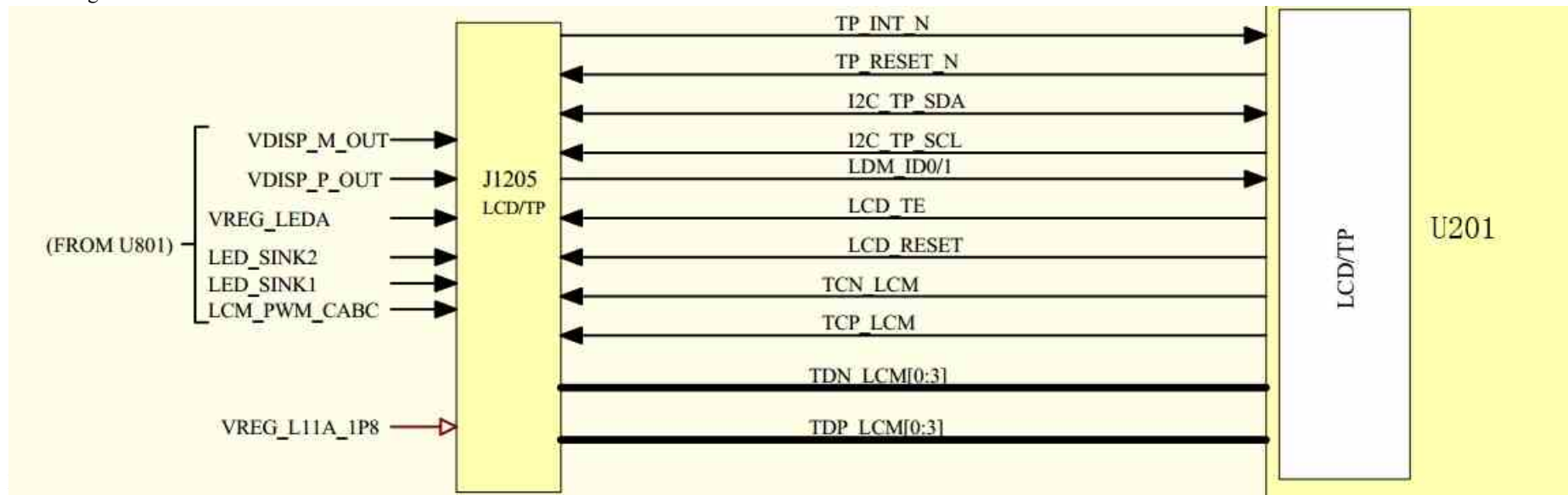
Fault Component:

U201

Maintenance analysis: The application crashes, after replacing U201, the fault repair.

3.9 Display function failure

U801 integrated LCD circuit, keyboard lamp circuit, TP circuit. Display, TP circuit principle block diagram:



显示测量表		
Symbol	测量值	测量点
VREG_L11A_1P8	1.8V	C1220
LDM_IDO	1.8V	R1201
LDM_ID1	1.8V	R1202
VREG_LED A	4-21V (点亮屏到21V)	B902
LED_SINK1	1.25V (点亮屏)	C1201
LED_SINK2	1.25V (点亮屏)	C1202
VDISP_M_OUT	-5.5	C931
VDISP_P_OUT	5.5V	C930

Display interface resistance reference

		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20			
		GND	420	GND	415	372	420	GND	470	∞	∞	490	420	420	GND	500	538	1	620	620	GND			
42	GND	J1205																				GND	43	
41																							44	
		GND	273	273	GND	273	273	GND	273	273	GND	273	273	GND	273	273	GND	1	590	GND	1			
		40	39	38	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21			

Maintenance analysis ideas: 1. Visually check whether J1205 and its peripheral components are damaged or falsely welded.

2. Software upgrade and troubleshooting.

3. Measure whether the ground value of each pin of J1205 is normal with multimeter diode file, and show whether the power supply is normal.

4. If the ground value is normal, measure whether the power supply and control signals in U801 meter are normal.

5. Replace U201.

Maintenance Case 1

Fault phenomenon: The brightness of the screen cannot be adjusted. Fault components: B903, L906

Maintenance analysis: The brightness of the screen cannot be adjusted, the test B903 is not conductive, and the fault is repaired after replacing B903 and L906.

Maintenance Case 2

Fault phenomenon: black screen

Faulty components: B902, B903

Maintenance analysis: Black screen, found B902, B903 burn marks, after replacement of the fault repair.

Maintenance Case 3

Fault phenomenon: black screen

Faulty components: B902, B903, L906

Maintenance analysis: Black screen, found that B902 and B903 have burnt marks, and still black screen after replacement. Measure that one end of L906 has 4V voltage, but the other end does not.

Repair the fault after replacing L906.

Maintenance Case 4

Fault phenomenon: Black screen, with backlight fault component: U801

Maintenance analysis: Black screen, backlight, detection DSP, DSN voltage is small, after replacing U801, fault repair.

3.10 Audio function failure

The audio circuit of Red Rice Note 5 includes speaker, microphone, earpiece and earphone. After the audio signal comes out of U801, it goes to different channels respectively. First, distinguish which part has the problem according to the fault phenomenon, and then analyze and repair according to the following modules.

3.10.1 Speaker function failure

Speaker is connected to the motherboard through FPC. Its principle is to go to CODEC through CPU first, and then amplify and output to interface J1303 through U1101 audio power amplifier and then to speaker. SPK meter:

扬声器测量值		
Symbol	测量值	测量点
VPH_PWR	4V	C1116
SPKR_OUT_P	工作状态最大到1.2V	R1101
SPKR_OUT_M	工作状态最大到1.2V	R1102
VREG_L13A_1P8	1.8V	C1117

Maintenance analysis ideas: 1.

Visual inspection of J1303 appearance is good.

2. Software upgrade and troubleshooting.

3. Measure whether SPKR_OUT_M and SPKR_OUT_P are normal to ground with multimeter diode gear.

4. Measure whether the working conditions of U1101 are normal, and replace U1101 abnormally.

5. Replace U201.

Maintenance Case 1

Fault Phenomenon:
Speaker Silent Fault
Component: Power Shield

Maintenance analysis: The speaker is silent, SPKR_OUT_M short circuit is measured, no short circuit is stopped after removing the power shield shell, and the fault is repaired after re-welding.

3.10.2 MIC function failure

Analysis ideas: 1. CIT test determines which MIC fault is, main MIC and motherboard MIC (MIC101)

2. Measure whether MIC power supply is normal and whether MIC_BIAS1 is normal.
3. Measure whether the primary and secondary MIC paths are normal. If it is the primary MIC, check whether the interface J1303 is abnormal; If this is a secondary MIC, replace UMIC101.
4. Replace U801.

3.10.3 Receiver function failure

Maintenance analysis ideas: 1. Whether the appearance inspection is normal.

2. Software upgrade and troubleshooting.
3. Measure whether the earpiece path is normal and whether the resistance value of REC1101 is abnormal.
4. Measure whether the power supply and enable signal of U801 is normal, and replace U801 if it is abnormal.
5. Replace U201.
- 6.

3.10.4 Headphone function failure

The earphone interface of Red Rice Note 5 is on the daughter board, which is connected to the motherboard through USB interface and controlled by U801. The detection of headphones is through CDC_HS_DET signal (this signal comes from U801). If this signal is abnormal, focus on checking U801.

Maintenance analysis ideas: 1. Check whether J1303 interface contacts are deformed or damaged.

2. Measure whether the CDC_HS_DET signal has a 1.8 V voltage.
3. Check the earphone channel according to the principle block diagram, and replace the faulty components on the corresponding circuit if there is any abnormality.
4. Replace U801.

3.11 Camera function failure

Redmi Note 5 rear camera is AI dual camera, 120 +5 million, Dual PD dual-core extremely fast focusing, front camera is 13 million, and has front flash.

Camera failures are divided into: rear main camera, rear sub-camera and front camera failures. The clock, reset and I2C signals are all separate, the camera is powered by different voltage regulators, and the enable signal is controlled by U201. Three cameras have one common power supply: VREG_L11A_1P8. If both front and rear cameras fail, this power supply can be considered first. If the front camera and rear main camera are normal and the rear auxiliary camera is faulty, SLAVECAM signal can be considered.

Camera interface measurements:



Maintenance analysis ideas: 1.

Software upgrade,
troubleshooting software.

2. Check whether J1204, J1203, J1202 and surrounding components are damaged.

3. Measure whether the ground values of J1204 and J1203/J1202 are normal.

4. Measure whether the camera power supply, clock and reset signal output are normal.

5. If all the above signals are normal, replace U201.

Maintenance Case 1

Failure phenomenon:

Camera jamming

Failure cause:

Software upgrade

Maintenance analysis: The camera is stuck, and the fault is repaired after the software upgrade

Maintenance Case 2

Fault Phenomenon:

Proactive Unable to

Open Fault Component:

EMI210

Maintenance analysis: The front camera can't be opened, and the ground value on the measured EMI210 side is abnormal, and the fault is repaired after replacement.

3.12 Touch screen function failure

The touch screen and display interface share J1205 and are controlled by U201. Maintenance analysis ideas: 1. Check whether the surrounding components are damaged.

2. Software upgrade and troubleshooting.

3. Measure whether the touch screen I2C, reset and interrupt signals are normal.

4. Replace U201.

Maintenance Case 1

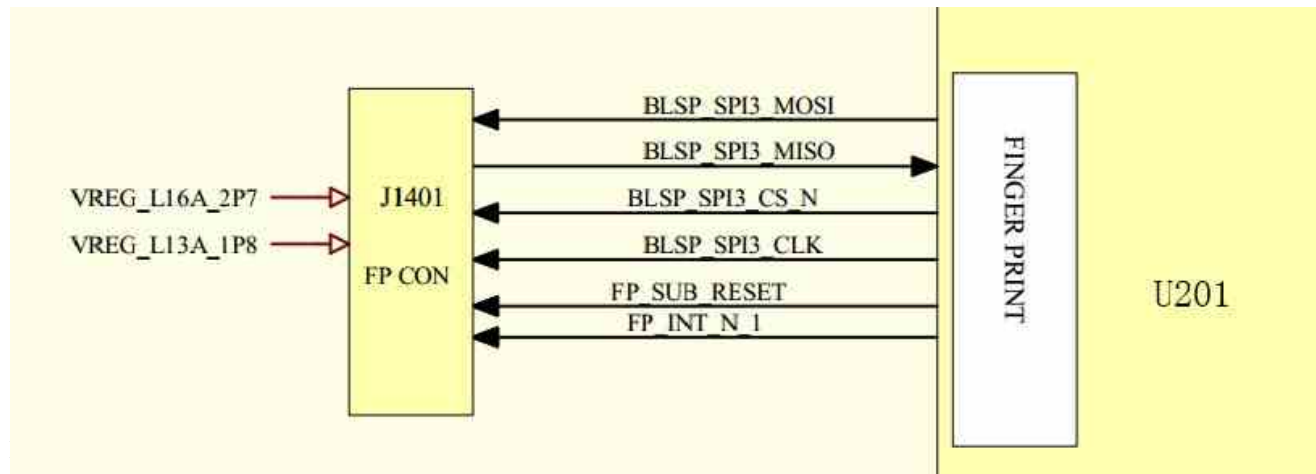
Failure phenomenon: touch screen failure

Faulty components: U201, R325

Maintenance analysis: touch screen failed, I2C_TP_SCL was measured for short circuit, and R326 was found to be short circuit, which was no longer short circuit after replacement, but R326 was close to one end of touch screen interface, and the voltage value was low, so the fault was repaired after CPU replacement.

3.14 Fingerprint recognition function failure

Fingerprint principle block diagram:



指纹测量表		
Symbol	测量值	测量点
VREG_L16A_2P7	2.7V	C1408
VREG_L13A_1P8	1.8V	C1402
FP_INT_N	0.9V	J1401第1PIN
FP_SUB_RESET	418	J1401第10PIN
BLSP_SPI3_MOSI	418	J1401第9PIN
BLSP_SPI3_MISO	418	J1401第8PIN
BLSP_SPI3_CS_N	420	J1401第7PIN
BLSP_SPI3_CLK	420	J1401第6PIN

Maintenance analysis ideas: 1.

Software upgrade,
troubleshooting software.

2. Measure whether the impedance of J401 to ground is normal.

3. Measure whether the voltage is normal.

4. Replace U201.

Maintenance Case 1

Failure Phenomenon:

Unable to identify
fingerprint Failure

Component: J1401

Maintenance analysis: Fingerprint can not be identified, observe the damage of motherboard fingerprint interface, and repair the fault after replacement.